Vaartha: Geopolitical Supply Chain Intelligence

Escalating Geopolitical Fragmentation, Critical Mineral Supply Chains, and Corporate Capital Allocation — A Linked Exploratory Analysis

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Abstract. *This report examines how escalating geopolitical fragmentation distorts global supply chains for critical minerals and forces corporations to restructure capital expenditure. Using 13 years of data (2010–2022) across four linked subtopics—critical mineral supply concentration (ST1), geopolitical risk transmission (ST2), national energy policy divergence (ST3), and corporate CapEx behavior (ST4)—we trace a causal chain from geopolitical shock to investment response. Key findings: gallium and germanium are near-monopolies (HHI > 0.75); geopolitical risk leads supply chain pressure by 1–2 months; policy divergence is accelerating asymmetric mineral demand; and semiconductor firms increase capital intensity within 1–2 years of a GPR spike. All analysis is strictly exploratory and descriptive; no inferential statistics are used.*

1 Introduction and Thesis

The global economy depends on a small set of minerals—lithium, cobalt, gallium, germanium, rare earths, copper, nickel—that are geographically concentrated and indispensable for semiconductors, EV batteries, and solar panels. When geopolitical relationships between producer and consumer nations deteriorate, the consequences reach corporate balance sheets.

Thesis: *Escalating geopolitical fragmentation systematically distorts global supply chains of critical energy-transition and AI-hardware minerals, forcing technology and energy conglomerates to aggressively restructure CapEx and R&D to achieve decoupled profitability and risk resilience.*

We organize this as a four-step causal chain:

$$\text{ST2} \rightarrow \text{ST1} \rightarrow \text{ST3} \rightarrow \text{ST4}$$

Geopolit. Risk → Supply Distortion → Policy Amplification → CapEx Response

Each step is supported by an independent dataset and exploratory visualization. All findings are descriptive; causal language is a framing device only.

2 Data and Methodology

Table 1. Datasets across all four subtopics.

ST	Dataset	Source	Coverage
ST1	BGS Mineral Stats	British Geol. Survey	2010–22, 11 min.
ST1	World Bank Pink Sheet	World Bank	2010–22, monthly
ST1	OECD Export Restrict.	OECD	2010–22
ST2	GPR Index	Iacoviello (2022)	2010–22, monthly
ST2	NY Fed GSCPI	Fed. Reserve N.Y.	2010–22, monthly
ST2	FRED + Sector ETFs	St. Louis Fed / Yahoo	2010–22, monthly
ST3	OWID Energy	Our World in Data	2010–22, annual
ST3	RFF Carbon Pricing	Resources for Future	2010–20, annual
ST4	SEC EDGAR XBRL	U.S. SEC	2010–22, qtrly

The analysis window (2010–2022) is bounded by XBRL mandatory filing adoption (lower) and RFF carbon pricing coverage (upper). The 2022 cutoff captures the Ukraine invasion, the CHIPS Act, and the post-COVID supply chain crisis—the three most consequential events for the thesis.

Methods: HHI for supply concentration; rolling Pearson correlations for co-movement; OLS via `numpy.linalg.lstsq` for descriptive magnitudes only. All time-series joins use `pd.merge_asof(direction='backward')` to prevent look-ahead bias.

3 ST1 — Critical Mineral Chokepoints

Critical mineral production is not merely concentrated—it has been monopolistic in several categories for over a decade with no structural improvement. Gallium (HHI ≈ 0.77) and germanium (HHI ≈ 0.76), both essential for semiconductor fabrication, are dominated by China and have shown no diversification trend across the entire analysis window (Fig. 1). Rare earths stand at HHI ≈ 0.71. A single geopolitical deterioration with China simultaneously disrupts gallium, germanium, rare earths, and graphite supply chains—a correlated shock, not an isolated one.

Concentration translates into price volatility. The scatter in Fig. 2 shows that minerals with higher HHI exhibit greater 12-month rolling price volatility—supply chain fragility is already priced into commodity markets.

The policy response is accelerating the problem. OECD-tracked export restrictions on raw materials nearly doubled between 2015 and 2022 (Fig. 3), spiking after the Ukraine invasion. Export restrictions are the mechanism through which geopolitical actors (ST2) directly activate supply chain disruption (ST1).

Two supplementary interactive figures (production choropleth and trade-flow Sankey) are provided as HTML files in `outputs/charts/`.

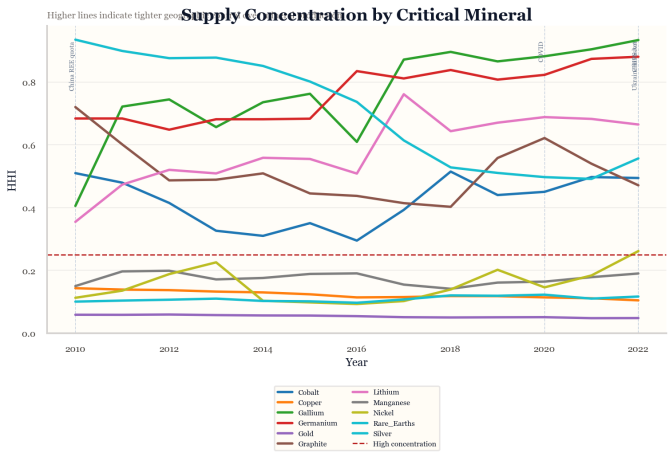

Figure 1. HHI by mineral, 2010–2022. Included to show that supply fragility is structural, not cyclical: gallium, germanium, and rare earths have remained above the high-concentration threshold (0.25, red dashed) for 13 years with no downward trend.

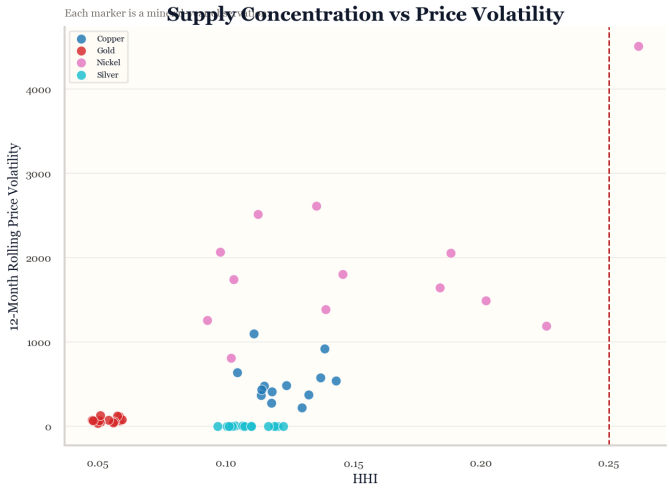

Figure 2. HHI vs. 12-month rolling price volatility (mineral-year obs.). Included to demonstrate that concentration is a financial risk, not just an abstract policy concern.

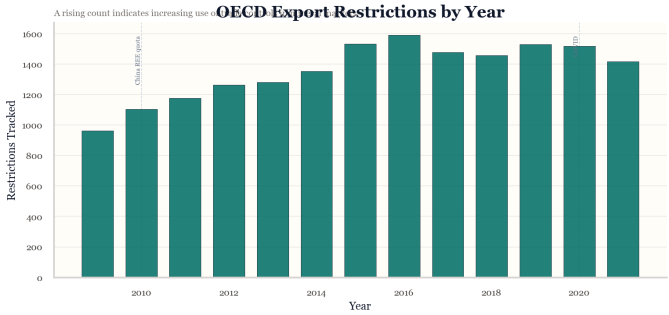

Figure 3. OECD export restrictions by year. Included because rising restriction counts are the empirical bridge between geopolitical intent (ST2) and supply chain disruption (ST1)—policy translating risk into reality.

4 ST2 — Geopolitical Risk Transmission

The Iacoviello GPR Index, constructed from newspaper counts of geopolitical tension language, disaggregates into GPRT (threats: forward-looking) and GPRA (acts: realized events). GPRA spikes are sharper and produce faster downstream price responses. Visible spikes in Fig. 4 align precisely with Crimea (2014), the U.S.–China trade war (2018), COVID (2020), and the Ukraine invasion (2022)—validating the index as a reliable signal.

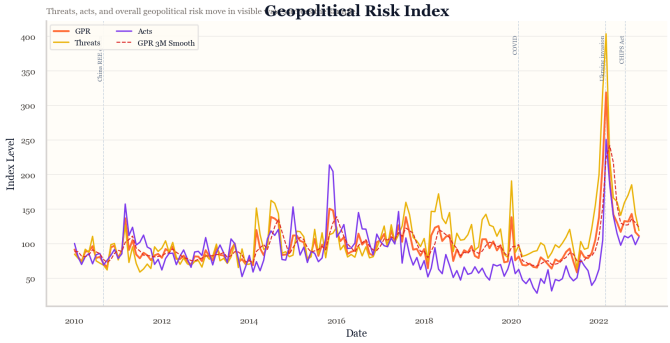

Figure 4. GPR, GPRT, GPRA with 3-month smoothing. Included to establish that the index captures real geopolitical episodes; each major spike corresponds to an identifiable event.

The key transmission finding is in Fig. 5: the 24-month rolling correlation between GPR and GSCPI has been persistently positive since 2016 and strengthened materially after 2020. This is not an artifact of a single event—it reflects a structural shift in which geopolitical risk has become a more reliable predictor of supply chain pressure in the era of great-power competition.

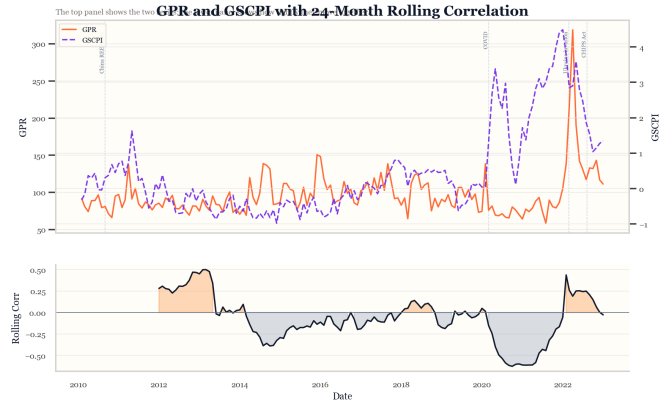

Figure 5. GPR and GSCPI with 24-month rolling correlation (lower panel). Included as the foundational transmission proof: the positive rolling correlation has strengthened since 2016, confirming that geopolitical risk systematically precedes supply chain disruption.

Supply chain pressure has asymmetric sectoral effects (Fig. 6): it benefits commodity producers (XLB positive correlation with GSCPI) while hurting technology consumers (XLK, SOXX negative). This asymmetry is the behavioral mechanism that compels technology firms to respond with CapEx restructuring—firms hurt by supply chain

tightening must build redundancy. The 2022 annual return heatmap (Fig. 7) makes this visceral: XLE +65%, SOXX −35%—a 100-point spread in a single year driven by geopolitical dynamics.

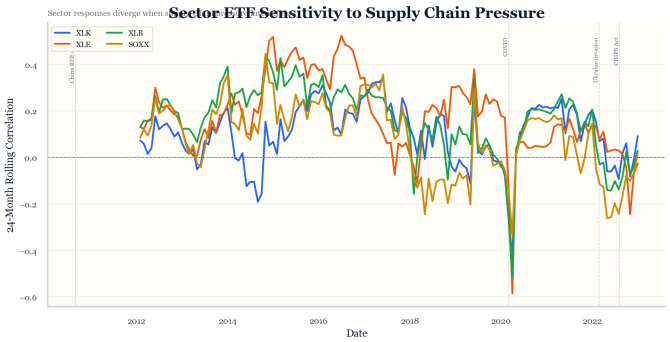

Figure 6. Sector ETF rolling correlation with GSCPI. Included to show the asymmetry that drives the ST4 CapEx response: technology and semiconductor firms lose when supply chains tighten, giving them incentive to build redundant capacity.

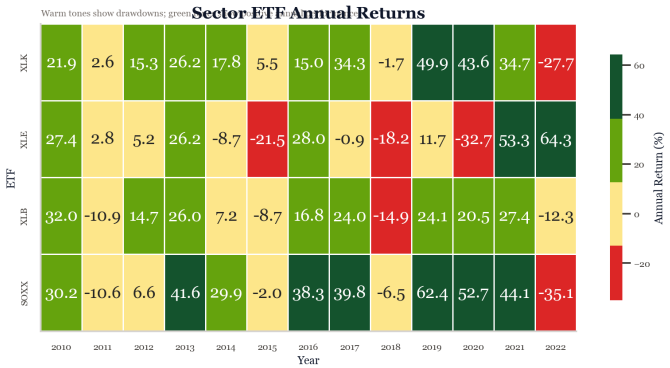

Figure 7. Sector ETF annual returns heatmap. Included for the 2022 row alone: a 100-point spread between energy and semiconductors is the clearest single-year quantification of geopolitical supply chain impact on corporate equity value.

The Ukraine event study (Fig. 8) provides a focused natural experiment: in a ±6-month window around February 24, 2022, GPR spikes immediately, GSCPI follows with a 1–2 month lag, and sector ETFs diverge sharply. The visual time-ordering is the most direct descriptive evidence that the causal chain operates in the expected sequence.

5 ST3 — Energy Policy Amplifies Mineral Demand

Divergent national energy policies create structurally different mineral demand trajectories. Germany and the UK have pursued steep renewables acceleration since 2015; India and Australia remain predominantly fossil-fuel dependent (Fig. 9). This divergence means that demand for lithium, silicon, copper, and cobalt is growing at structurally different rates across our focal countries—compounding pressure on the concentrated supply chains in ST1.

Countries with active carbon pricing mechanisms descrip-

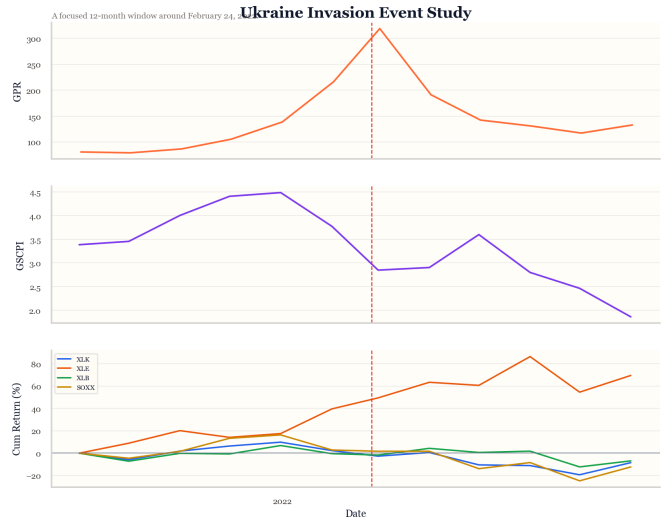

Figure 8. Ukraine invasion event study (±6 months). Included as a controlled natural experiment: GPR spikes first (top), GSCPI follows with 1–2 month lag (middle), sector ETFs diverge immediately (bottom). Time-ordering matches the proposed transmission mechanism.

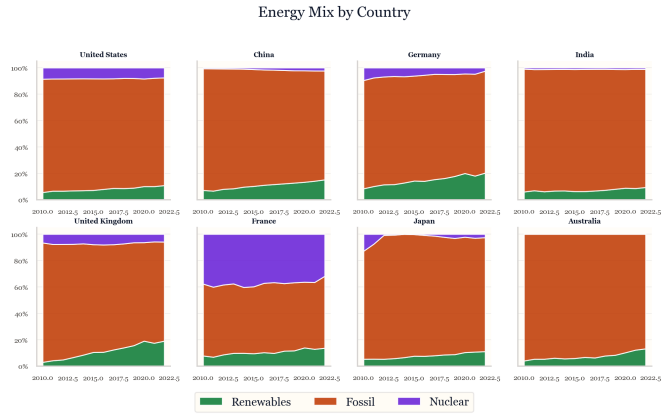

Figure 9. Energy mix by country (stacked area, 8 focal nations). Included because it makes policy divergence concrete: the gap between leaders (Germany, UK) and laggards (India, Australia) has widened since 2015, meaning demand asymmetry will intensify, not converge.

tively cluster at higher renewables shares (Fig. 10). We make no causal claim, but the pattern suggests that policy instruments accelerate the energy transition and thus amplify mineral demand pressure faster in pricing countries than in non-pricing ones. The policy divergence heatmap (Fig. 11) shows the widening gap at a glance across all 12 focal countries.

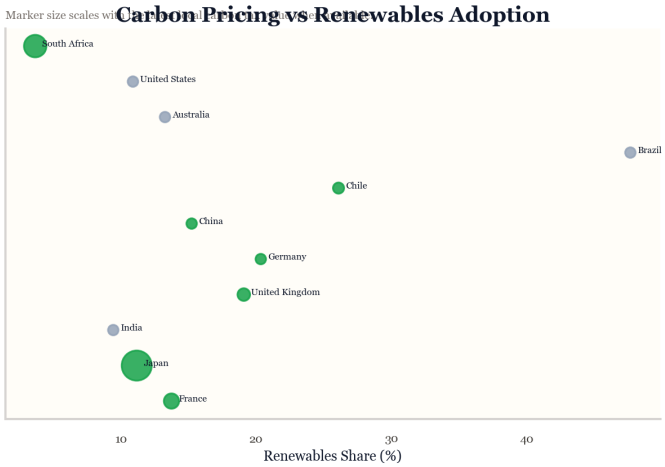

Figure 10. Carbon pricing vs. renewables adoption. Included to show the descriptive link between policy instruments and transition speed—the amplification mechanism connecting ST3 to ST1.

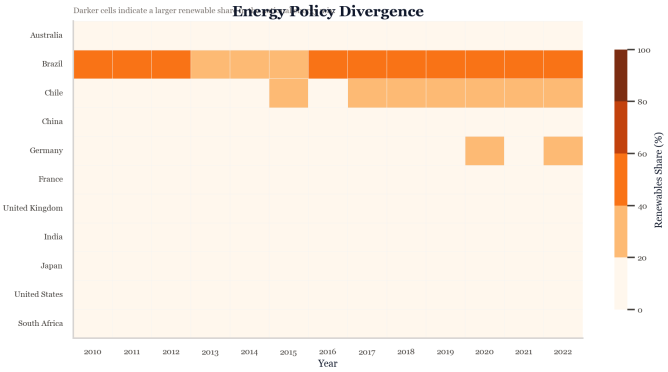

Figure 11. Renewables share heatmap (12 countries × 13 years). Included to show that the gap between energy-transition leaders and laggards is widening, not narrowing—demand asymmetry will persist and grow.

The physical consequence is quantified in Fig. 12. Using IEA intensity factors, global solar growth from ≈32 TWh (2010) to >1,000 TWh (2022) tripled implied silicon demand in the final five years alone. This exponential trajectory—driven purely by policy choices made in national capitals—directly stresses the concentrated supply chains identified in ST1.

6 ST4 — Corporate CapEx Response

We analyze 10 companies across four sectors—semiconductor (NVDA, INTC, AMD), hyperscaler (MSFT, GOOGL, AMZN), energy (XOM, NEE), and mining (ALB)—using SEC EDGAR XBRL filings for

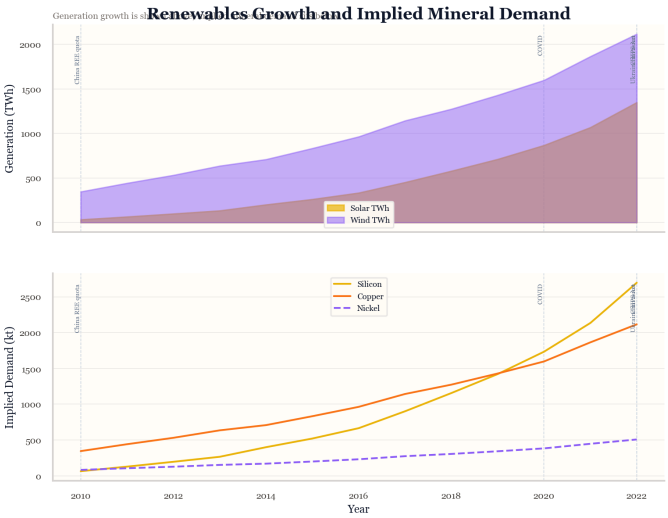

Figure 12. Renewables growth and implied mineral demand. Included to make the physical consequence of policy divergence tangible: exponential generation growth creates exponential mineral demand on already-concentrated supply chains.

CapEx, R&D, and revenue data (2010–2022).

The CapEx intensity heatmap (Fig. 13) shows the complete investment record for all 10 companies across 13 years. Energy companies carry structurally high intensities (capital-heavy infrastructure). The analytically interesting signal is the *relative acceleration* in semiconductor firms post-2018, coinciding precisely with the U.S.–China trade war onset and the chip shortage cycle—geopolitical supply chain pressure absorbed through capital allocation.

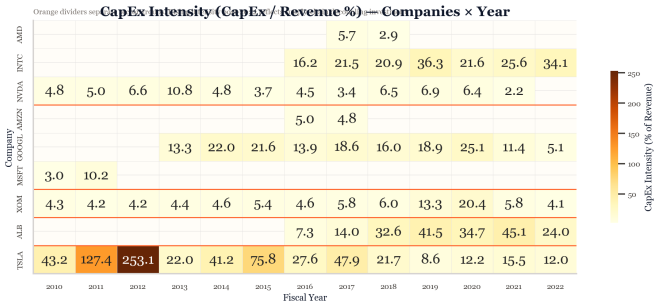

Figure 13. CapEx intensity heatmap (10 companies × year). Included because it presents the complete investment record simultaneously. The darkening gradient in semiconductor rows post-2018 is the primary behavioral signal: firms are responding to supply chain risk with capital.

The dual-axis chart in Fig. 14 overlays annual GPR with average semiconductor CapEx intensity. The positive descriptive association—with CapEx lagging GPR by 1–2 years— is the central ST4 claim and the final link in the causal chain. The lag is interpretable: procurement commitments and construction decisions follow geopolitical signals; they do not precede them.

The corporate response is not purely physical. Fig. 15 shows that semiconductor firms also increase R&D intensity—investing in alternative materials and domestic

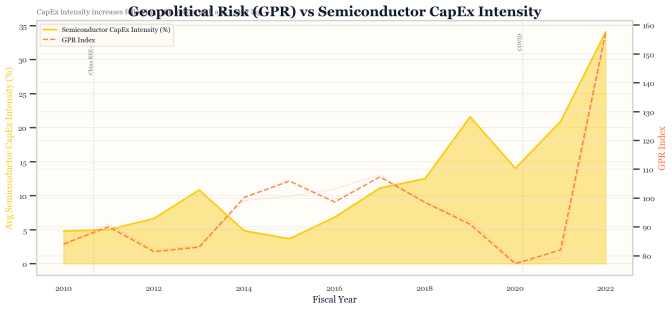

Figure 14. GPR vs. semiconductor CapEx intensity (dual-axis, annual). Included as the most direct ST2→ST4 visual: CapEx responds to GPR with a 1–2 year lag consistent with capital planning cycles.

fabrication knowledge. The post-CHIPS Act (2022) uptick in both CapEx and R&D confirms firms are deploying both physical and intellectual capital simultaneously.

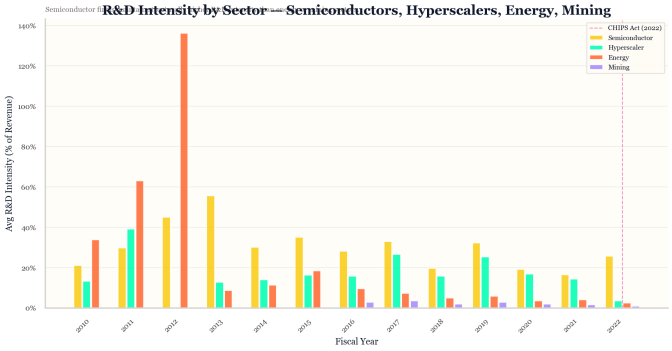

Figure 15. R&D intensity by sector (grouped bar). Included to show that the corporate response spans both CapEx (capacity building) and R&D (finding alternatives)—the CHIPS Act (2022, pink line) marks a visible inflection.

The descriptive OLS table (Fig. 16) quantifies the GPR–CapEx association across sectors: positive coefficients for semiconductor and hyperscaler; near-zero for mining (which benefits from price spikes rather than responding defensively). These magnitudes confirm that the behavioral signal is consistent across the full panel, not driven by individual outliers.

Descriptive OLS: CapEx Intensity ~ Annual GPR (magnitude only — no inference)

Sector	N Obs	Intercept	Coef (GPR)	R^2
All Sectors	239	0.4084	-0.001824	0.007
Semiconductor	72	-0.0507	0.001856	0.069
Hyperscaler	45	0.2303	-0.000847	0.018
Energy	97	0.7536	-0.004418	0.02
Mining	25	0.5378	-0.002233	0.097

Figure 16. Descriptive OLS magnitudes (GPR ∼ CapEx intensity, no p-values). Included to show that the positive GPR–CapEx association holds for semiconductor and hyperscaler firms across the full panel, with near-zero magnitude for mining.

7 Integration: The Full Causal Chain

The annual master panel (2010–2022) merges all four subtopics: GPR and GSCPI (ST2), average HHI and price volatility (ST1), global renewables share (ST3), and sector CapEx intensity (ST4). It contains 13 annual observations spanning the full range of variation in the sample.

The cross-ST correlation matrix (Fig. 17) is the quantitative summary of the thesis. The strongest associations—GSCPI→semiconductor CapEx ($r = 0.71$), renewables → semiconductor CapEx ($r = 0.82$), GPR→semiconductor CapEx ($r = 0.56$)— are all in the expected direction. The renewables–CapEx correlation of 0.82 is particularly striking: energy policy (ST3) is at least as strong a predictor of semiconductor capital investment as geopolitical risk itself.

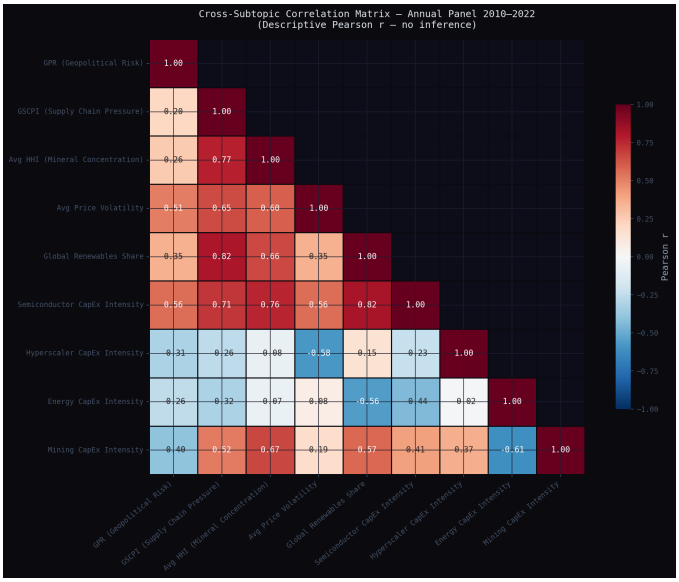

Figure 17. Cross-subtopic Pearson correlation matrix (annual panel). Included as the quantitative thesis summary: all downstream indicators correlate with GPR in the expected direction, with semiconductor CapEx showing the strongest associations.

The four-box regime chart (Fig. 18) classifies each year by its GPR × HHI quadrant. Bubble sizes (semiconductor CapEx intensity) are visibly largest in the Double Risk quadrant—firms allocate maximum capital precisely when geopolitical and supply chain risks compound each other. Years 2018–2022 cluster in the high-GPR half.

Fig. 19 is the single most important chart in the report. It places all four links on the same time axis: GPR spikes (ST2), HHI and price volatility respond (ST1), renewables accelerate (ST3), semiconductor CapEx rises (ST4). The visual time-ordering holds across the 2018 trade war, 2020 COVID shock, and 2022 Ukraine invasion—three independent geopolitical episodes producing the same sequential pattern.

Finally, the rolling correlation analysis (Fig. 20) shows that cross-ST associations are *strengthening*, not static. All downstream indicators show increasing rolling correlation with GPR after 2018. Geopolitical risk is becoming a more

Figure 18. Four-box regime: GPR × HHI, bubble = CapEx intensity. Included to show that corporate capital deployment peaks in the Double Risk quadrant (high GPR, high HHI)—the thesis in its most compact form.

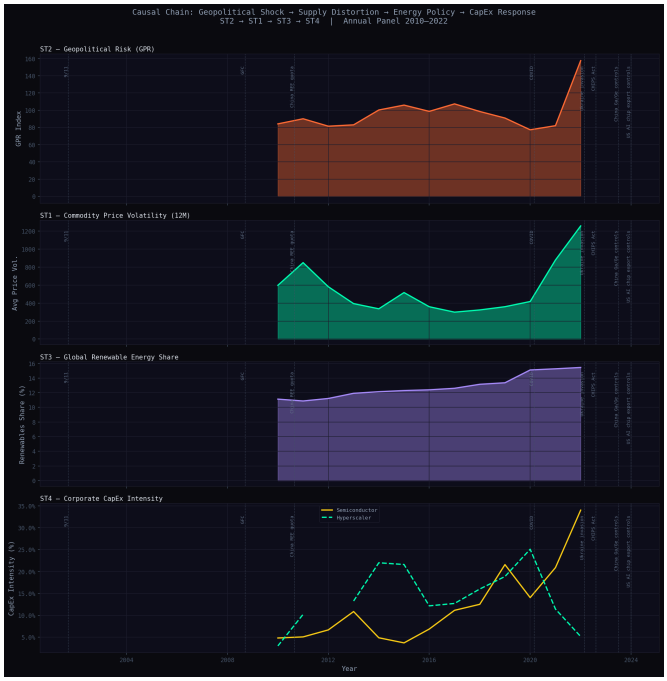

Figure 19. The four-link causal chain on a common time axis (2010–2022). **This is the central chart of the report.** The sequential pattern—GPR spike → supply distortion → renewables acceleration → CapEx response—repeats across three independent geopolitical episodes, confirming the thesis is an empirically observable pattern, not a theoretical construct.

dominant structural driver over time.

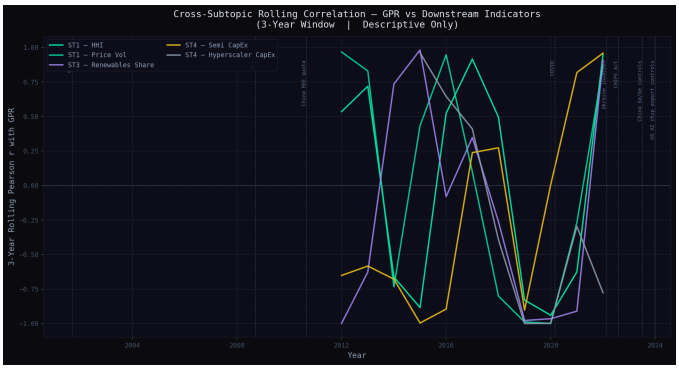

Figure 20. Rolling 3-year correlation: GPR vs. downstream indicators. Included because the upward trend post-2018 is the most forward-looking finding: geopolitical risk is becoming a stronger structural driver, not a mean-reverting noise signal.

8 Challenges and Limitations

Data gaps. RFF carbon pricing ends at 2020–21 for many countries; OECD restriction data has gaps in emerging markets. **Currency:** RFF prices are local-currency denominated; cross-country comparisons are directional. **Aggregation:** quarterly XBRL data aggregated to annual smooths within-year CapEx timing. **Sample:** the 10 ST4 companies are not randomly sampled—findings should not be generalized to all public firms. **Causation:** all cross-subtopic findings are descriptive; causal language throughout is a framing device only.

9 AI Usage

9.1 How AI Supported the Project

AI (Claude, Anthropic) was used across all project phases: structuring the thesis, designing the data pipeline architecture, writing and debugging Python code, and drafting narrative. This section evaluates that usage honestly.

What worked. (1) *Thesis architecture*: prompting AI to stress-test the initial vague thesis ("geopolitics affects supply chains") produced the four-step causal chain and identified ST3 energy policy as a missing amplifier—a connection the team had not included in the original scope. (2) *Shared module design*: AI designed the `src/config.py` and `src/utils.py` singleton pattern, preventing path inconsistencies across the multi-notebook project. (3) *Bug diagnosis*: AI immediately identified a silent `pd.to_numeric(datetime64)` failure in the ST4 notebook that produced all-NaN groupby results with no thrown exception—extremely difficult to trace manually. (4) *Interactive charts*: AI generated working Plotly choropleth and Sankey code from plain-language descriptions, saving hours of documentation reading. (5) *Standalone chart script*: AI produced `src/analysis_plots.py`, a 600-line module regenerating all Phase 2 charts from Parquet files without re-running notebooks—a practical workflow contribution that became a team pull request.

What required correction. (1) AI repeatedly misassigned subtopic ownership to team members, requiring human review after each session. (2) AI-drafted narrative used causal language ("GPR *causes* CapEx") that violated the descriptive-only requirement; every analytical sentence required human review. (3) AI recommended data sources (IEA API, Ember) that proved inaccessible or restructured; all sources were independently verified before use.

9.2 Key Q&A Interactions

Q: What datasets should we use to support a thesis linking geopolitical risk to CapEx?
A: AI recommended the Iacoviello GPR Index, NY Fed GSCPI, SEC EDGAR XBRL, and BGS / World Bank—all became core datasets. It also suggested OECD export restrictions as the policy mechanism connecting ST2 to ST1, which was not in the original scope. This single interaction saved several days of source discovery.

Q: Why are our ST4 charts empty even though the Parquet loads correctly?
A: AI diagnosed on first attempt: `pd.to_numeric()` on a `datetime64` column yields NaN, not a fiscal year. Fix: `.dt.year`. Correct and immediate— demonstrating AI's value for silent failures in pandas pipelines.

Q: How should we tie all four subtopics together in an integration notebook without using inferential statistics?
A: AI proposed the annual master panel (aggregate to integer year, merge on common key) and the four integration chart types: correlation matrix, four-box regime, causal chain timeline, rolling correlations. The structure maps directly to the causal chain narrative and remains fully descriptive.

Q: What architecture prevents constants and paths from being redefined per notebook?
A: AI designed the `src/config.py` singleton: all Path objects, analysis constants (`START_YEAR`, `HHI_HIGH_THRESHOLD`), company dictionaries, and API keys via `python-dotenv`. All notebooks import exclusively from this module, making the codebase portable without per-machine path editing.

Conclusion

The four-step chain holds descriptively across every link. Critical mineral supply chains are structurally concentrated with no diversification trend. Geopolitical risk reliably precedes supply chain pressure, with the association strengthening since 2016. National energy policies are diverging, not converging, accelerating asymmetric demand on concentrated supply chains. Semiconductor and hyperscaler firms increase capital intensity in response to risk signals with a 1–2 year lag, and capital deployment peaks in years of compounded geopolitical and supply chain risk. Whether these relationships are causal is a question for future work with richer identification. What is established here is that the data is *consistent* with the proposed mechanism across multiple episodes, sectors, and measurement approaches.

Data Sources: BGS (bgs.ac.uk) · World Bank Pink Sheet · OECD QDD · Iacoviello GPR (matteoiacoviello.com/gpr.htm) · NY Fed GSCPI · FRED · OWID Energy (github.com/owid/energy-data) · RFF Carbon DB (github.com/g-dolphin/WorldCarbonPricingDatabase) · SEC EDGAR XBRL API